

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fourth Semester B.Tech (CBCS) Degree Examinations

Summer Semester July 2024

EC3003-1 – DIGITAL SIGNAL PROCESSING (EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from Unit – I & Unit – II each and **One full** question from Unit – III.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The 4-point discrete Fourier Transform (DFT) of a discrete time sequence $\{1, 0, 2, 3\}$ is
 - $[2, 2+2j, 6, 2-2j]$
 - $[6, -1+3j, 0, -1-3j]$
 - $[6, 1-3j, 2, 1+3j]$
 - $[0, 2+2j, 2, -2-2j]$
- The first six points of the 8-point DFT of a real valued sequence are 5, $1-j3$, 0, $3-4j$, 0 and $3+j4$. The last two points of the DFT are respectively
 - 0, $1-j3$
 - 0, $1+j3$
 - $1+j3$, 5
 - $1-j3$, 5
- If $W_4^{100} = W_x^{200}$ then what is the value of x?
 - 16
 - 4
 - 8
 - 2
- The value of the phase factor W_8^3 is
 - $0.707-0.707i$
 - $-0.707-0.707i$
 - $0.707+0.707i$
 - $-i$
- The number of complex multiplications required for 500-point DFT computation using FFT (approximately) is
 - $500 * 500$
 - $500 * 499$
 - $400 * 400$
 - $499 * 200$
- FFT is a modification of DFT in terms of
 - Noise
 - Computational Speed
 - Linearity
 - None of these
- In an N point sequence, if $N=16$, the total number of complex additions and multiplications using Radix-2 FFT are
 - 64, 80
 - 24, 12
 - 240, 256
 - 80, 64
- In 8-point DFT computation, by using radix-2 FFT, there are stages of computations with butterflies per stage.
 - 2, 3
 - 5, 3
 - 3, 4
 - 4, 5
- Length of the impulse response of a IIR filter is ____
 - Finite
 - None of these
 - zero
 - Infinite

10. A system where the pass band cutoff frequency is greater than stop band cutoff frequency is
- A) Low pass filter B) High pass filter
C) Band pass filter D) Band reject filter
11. Filter design that plots all the poles of the filter on the unit circle is
- A) Chebyshev Filters B) FIR Filters
C) Butterworth Filters D) Elliptic Filters
12. Which of the following methods are used to convert analog filter into digital filter?
- A) Impulse invariant technique B) Convolution technique
C) Frequency sampling technique D) Window technique
13. Consider the assertions given below. Which among them is an advantage of FIR Filter?
- A) Necessity of computational techniques for filter implementation B) Requirement of large storage
C) Incapability of simulating prototype analog filters D) Presence of linear phase response
14. The values of cutoff frequencies in general depend on which of the following?
- A) Type of the window B) Length of the window
C) Type & Length of the window D) None of the mentioned
15. In frequency sampling method, transition band is a multiple of which of the following?
- A) π/M B) $2\pi/M$
C) $\pi/2M$ D) $2\pi M$
16. Tapering in window result in lower side lobes at the expense of an increase in the width of the _____ of the filter.
- A) Transition Band B) Pass Band
C) Stop Band D) All of these
17. The pipelining refers to
- A) Prefetching instructions and storing in a FIFO queue B) Fetching instruction and data simultaneously
C) Executing different phases D) Executing different instruction in parallel using two or more computational units
18. The format is commonly used in Modern DSP processors
- A) Fixed Point B) Floating Point
C) IEEE Floating Point D) None of these
19. In DSP, in order to preserve the computational speed of single-cycle instruction execution, shifts by several bits should be accomplished in a single cycle. This is possible by a combinational circuit known as
- A) Braun Multiplier B) Barrel Shifter
C) ALU D) MAC
20. Fixed point filter implementation uses
- A) 2's complement integer format B) Q point format
C) Single Precision point D) Double precision point

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Find the DFT of the sequence, $x(n) = \begin{cases} 1; 0 \leq n \leq 2 \\ 0; \text{otherwise} \end{cases}$ For N=4 & N=8. Also sketch magnitude and phase spectra.	8	L*3	1	1
b)	Find the circular convolution of the following two sequences using concentric circles method. $x_1(n) = \{2,1,2,1\}$ $x_2(n) = \{1,2,3,4\}$	8	L3	1	1

2. a) Consider a FIR filter with input sequence $x(n) = \{1, 2, 3, 3, 2, 1, -1, -2, -3, 5, 6, -1, 2, 0, 2, 1\}$ and impulse response $h(n) = \{3, 2, 1, 1\}$. Find the output $y(n)$. Use overlap add method assuming the length of the block as 7. 8 L3 2 1
- b) Find DFT of the sequence $x(n) = \{1, 2, 4, 8, 16, 32, 64, 128\}$ using DIT FFT algorithm. 8 L3 2 1
3. a) Given a sequence $x(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$. Find $X(K)$ using DIFFFT algorithm. 8 L3 2 1
- b) Find $x(n)$ using inverse DIFFFT algorithm.
 $X(k) = \{36, -4 + 9.656i, -4 + 4i, -4 + 1.656i, -4, -4 - 1.656i, -4 - 4i, -4 - 9.656i\}$ 8 L3 2 1

Unit – II

4. a) Design a Chebyshev lowpass filter for the following specifications.
 (i) Acceptable passband ripple of 2 dB
 (ii) Passband edge frequency of 1 rad/sec
 (iii) Stopband attenuation of 20 dB or more at 1.3 rad/sec 8 L3 3 1
- b) With neat characteristics, derive expressions for order and cut off frequency of a Butterworth low pass filter. 8 L3 3 1
5. a) Design an analog Butterworth low pass filter to meet the following specifications.
 Pass band gain = -1 dB
 Pass band edge frequency = 4 rad/sec
 Stop band attenuation greater than equal to 20 dB
 Stop band edge frequency = 8 rad/sec 8 L3 3 1
- b) Design a 7 tap linear phase FIR filter with a cutoff frequency $\pi/2$. The design is to be done based on frequency sampling technique. Also, find the filter coefficients. 8 L3 4 1
6. a) The desired frequency response of a low pass filter is given by
- $$H_d(\omega) = \begin{cases} e^{-j2\omega}, & |\omega| \leq \pi/4 \\ 0; & \pi/4 \leq |\omega| \leq \pi \end{cases}$$
- Determine the frequency response of the FIR filter if a rectangular window is used. 8 L3 4 1
- b) Discuss with necessary equations and sketches, bilinear transformation technique for converting analog transfer function to digital transfer function. 8 L3 3 1

Unit – III

7. a) Discuss the following DSP Architectural Units with necessary diagrams.
 (i) MAC unit
 (ii) ALU
 (iii) Barrel Shifter 8 L2 5 1
- b) Discuss the following with necessary diagrams.
 (i) Von Neumann Architecture & Harvard Architecture
 (ii) Fixed point and Floating point formats 8 L2 5 1
8. a) Discuss the following addressing modes with examples and sketches.
 (i) Circular addressing mode
 (ii) Bit Reversed addressing mode. 8 L2 5 1

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- b) (i) Find the Q-15 representation for the decimal number -0.160123.
(ii) Convert the Q-15 signed number 0.100011110110010 to the decimal number.
(iii) Add the two numbers 0.100011110110010 and 1.110101110000010.

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
